Throughout, $\Bbbk$ is a field, and all modules are right modules unless a left action is specified.

Exercise 1. Graded algebras, duals, and non-graded modules

(1) For $A = \Bbbk A_n$ with $n \geq 2$, verify that $A_d = \operatorname{span}_\Bbbk\{e_{ij} \mid j - i = d\}$ defines a grading on A.

(2) Compute $\operatorname{rad}(\Bbbk A_n)$ from its definition as the intersection of all maximal right ideals.

(3) Let $A = \Bbbk A_3$, $M = e_{22}A$, and consider $\mathbb{D}(M)_d = \operatorname{Hom}_\Bbbk(M_{-d}, \Bbbk)$ with $(af)(m) = f(ma)$.

i. Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of matrix units on it.

ii. Show that $\Bbbk e_{13} \subset Ae_{33}$ is a graded left submodule and that $\mathbb{D}(M) \cong (Ae_{33}/\Bbbk e_{13})(1)$ as graded left A-modules.

iii. Deduce that $\mathbb{D}(M)$ is not isomorphic to any $(Ae_{jj})(r)$, $j \in \{1, 2, 3\}$, $r \in \mathbb{Z}$.

(4) **Modules that are not graded.** Let A be the algebra of matrices

$$X = \begin{pmatrix} a & u\alpha + v\beta \\ 0 & b \end{pmatrix} \qquad (a, b, u, v \in \Bbbk),$$

with the usual matrix product, where scalars act on the space $\Bbbk\alpha \oplus \Bbbk\beta$ in the obvious way:

$$\begin{pmatrix} a & p \\ 0 & b \end{pmatrix}\begin{pmatrix} a' & p' \\ 0 & b' \end{pmatrix} = \begin{pmatrix} aa' & ap' + pb' \\ 0 & bb' \end{pmatrix} \qquad (p, p' \in \Bbbk\alpha \oplus \Bbbk\beta).$$

Grade A by $\deg\alpha = 1$, $\deg\beta = 0$, with diagonal entries in degree 0. For $(\lambda, \mu) \in \Bbbk^2$ let $M(\lambda, \mu) = \Bbbk^2$ (row vectors) with right action

$$(x_1, x_2) \cdot X = (ax_1,\ (\lambda u + \mu v)x_1 + bx_2).$$

i. Verify that A is a graded algebra and that $M(\lambda, \mu)$ is a right A-module for every $(\lambda, \mu) \in \Bbbk^2$.

ii. Prove that $M(\lambda, \mu)$ admits a grading if and only if $\lambda\mu = 0$.

(5) **Independence of the symmetrising form.** Let $A = \Bbbk A_2$, $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$ and $M = e_{11}A$ (inherited gradings). For $c = (c_1, c_2) \in (\Bbbk \setminus \{0\})^2$, consider the symmetrising form

$$\tau_c(ae_{11} + be_{22}) = c_1 a + c_2 b.$$

Recall that we have the natural isomorphism $\alpha_c : (-)^{(*)} \xrightarrow{\sim} \mathbb{D}$ associated to τ_c given by $f \mapsto \tau_c \circ f$, which induces a left A-action $\cdot_c$ on $M^{(*)}$ induced by α_c is given by

$$a \cdot_c f = \alpha_c^{-1}\big(a\,\alpha_c(f)\big), \qquad \text{where } (a\lambda)(m) = \lambda(ma) \text{ for } \lambda \in \mathbb{D}M.$$

Let

$$\varphi_0(ae_{11} + be_{12}) = ae_{11}, \qquad \varphi_1(ae_{11} + be_{12}) = be_{22}.$$

i. Check that φ_0, φ_1 forms a homogeneous basis of $M^{(*)}$ (specify their degrees). Compute $e_{11} \cdot_c \varphi_i$, $e_{22} \cdot_c \varphi_i$, $e_{12} \cdot_c \varphi_i$ for $i \in \{0, 1\}$. Prove that $e \cdot_c \varphi_i = e \cdot_{c'} \varphi_i$ for all $e \in \{e_{11}, e_{22}, e_{12}\}$ and $i \in \{0, 1\}$ if and only if $(c_1', c_2') = \rho(c_1, c_2)$ for some $\rho \in \Bbbk \setminus \{0\}$.

ii. More generally, let A be a locally finite-dimensional graded algebra with semisimple $A_0 = S$, and τ, τ' be symmetrising forms on S with associated natural isomorphisms α_τ and left A-action $\cdot_\tau$. Prove that there is an invertible $z \in S$ with $\tau'(s) = \tau(zs)$ for all s, and that

$$\gamma : (M^{(*)}, \cdot_{\tau'}) \to (M^{(*)}, \cdot_\tau), \quad (\gamma f)(m) = z\, f(m),$$

is an isomorphism of graded left A-modules. Deduce that in (i) the two modules $(M^{(*)}, \cdot_c)$ and $(M^{(*)}, \cdot_{c'})$ are isomorphic for all c, c'.

Hint. Compare $\alpha_{\tau'}$ with α_τ.

Exercise 2. *S-duals and quadratic duals*
Let S be a finite-dimensional semisimple $\Bbbk$-algebra. Let V be a finite-dimensional right S-module and W a finite-dimensional S-bimodule, and define

$$\theta_r^{V,W} : W^* \otimes_S V^* \longrightarrow (V \otimes_S W)^*, \qquad \theta_r(g \otimes f)(v \otimes w) = g(f(v)w).$$

(1) Prove that θ_r is well defined S-module isomorphism.

(2) Let $p : V \to V'$ be right S-linear and $q : W \to W'$ a bimodule map. Prove

$$(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*).$$

(3) Let

$$S = \Bbbk^{\oplus 3}, \qquad V = \Bbbk a, \quad W = \Bbbk c, \qquad a \in e_1 V e_2, \quad c \in e_2 W e_3.$$

Compute $W^* \otimes_S V^*$, $V^* \otimes_S W^*$, $(V \otimes_S W)^*$ and describe θ_r.

(4) Let

$$S = \Bbbk[x]/(x^2), \qquad V = W = S/(x).$$

Show that

$$\dim_{\Bbbk}(W^* \otimes_S V^*) = \dim_{\Bbbk}(V \otimes_S W)^* = 1, \qquad \theta_r = 0.$$

At which step of your proof of (2) does the argument fail?

(5) Let $S = \bigoplus_{i=1}^4 \Bbbk e_i$ be a semisimple algebra with coordinate idempotent e_i for $1 \le i \le 4$ and let $V = \Bbbk a \oplus \Bbbk b \oplus \Bbbk c \oplus \Bbbk d$ be the S-bimodule with

$$a \in e_1 V e_2, \qquad b \in e_2 V e_4, \qquad c \in e_1 V e_3, \qquad d \in e_3 V e_4,$$

and $e_i V e_j = 0$ for all other (i, j). For $\lambda \in \Bbbk$ put

$$A_\lambda = T_S(V)/(ab - \lambda cd).$$

i. Using θ_r, find $R^\perp$ and a presentation of $A_\lambda^!$, for every $\lambda \in \Bbbk$.
Hint. Evaluate $\mu\, b^* a^* + \nu\, d^* c^*$ on $ab - \lambda cd$.

ii. Dualise once more: compute ${}^!(A_\lambda^!)$ and recover the relation $ab - \lambda cd$.

iii. Prove that $A_\lambda \cong A_\mu$ as graded algebras if and only if λ, μ are either both zero or both non-zero.
Hint. A graded isomorphism over S sends each of a, b, c, d to a scalar multiple of itself.

Exercise 3. Quadratic presentations and Hilbert series

(1) For $n \geq 3$ and $Z \subset \{1, \dots, n-2\}$ let

$$A_Z = \Bbbk A_n / (e_{k,k+2} \mid k \in Z)$$

with the quotient grading.

i. Show that

$$\{e_{ii}\}_i \ \cup \ \{e_{ij} \mid i < j, \ \nexists\, k \in Z : \ i \leq k \leq j-2\}$$

is a basis of A_Z.

ii. Write $A_Z = T_S(V)/(R)$ with explicit S, V, and $R \subset V \otimes_S V$.

(2) Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ and $V = \Bbbk a \oplus \Bbbk b$ with

$$a \in e_1 V e_2, \qquad b \in e_2 V e_1,$$

and $e_i V e_j = 0$ otherwise.

i. Find a basis of $T_S(V)_d$ for all $d \geq 0$, and compute $h_{T_S(V)}(t)$.
Hint. Count alternating words in each degree, including degree 0.

ii. Find a basis of $A = T_S(V)/(ab)$ and show that $\dim_{\Bbbk} A < \infty$.

(3) Let

$$B = T_{\Bbbk}(\Bbbk x \oplus \Bbbk y)/(x^2), \qquad \deg x = \deg y = 1, \qquad a_d := \dim_{\Bbbk} B_d.$$

i. Find a basis of B_d and compute $a_0, \dots, a_6$.

ii. Prove that

$$a_d = a_{d-1} + a_{d-2} \quad (d \geq 2), \qquad h_B(t) = \frac{1+t}{1-t-t^2}.$$

Hint. Sort the words by their first letter.

iii. Compute $B^!$ using θ_r (with $X = x^*$, $Y = y^*$), find a homogeneous basis of $B^!$ and $h_{B^!}(t)$, and verify

$$h_B(t)\, h_{B^!}(-t) = 1.$$

(1) Let $S = \bigoplus_{i=1}^{r} \Bbbk e_i$, let V be a finite-dimensional S-bimodule in degree 1, and put

$$C_{ij} = \dim_{\Bbbk} e_i V e_j, \qquad A = T_S(V)/(V \otimes_S V) = S \oplus V.$$

i. Prove that

$$A^! = T_S(V^*), \qquad h_A(t) = I + tC, \qquad h_{A^!}(t) = (I - tC^{\mathsf{T}})^{-1}.$$

Hint. Use $V \otimes_S V$ as the relation space and $h_{T_S(V)}(t) = (I - tC)^{-1}$ (Lecture 2).

ii. Deduce that

$$h_A(t)\, h_{A^!}(-t)^{\mathsf{T}} = I.$$

(2) Let A be a locally finite-dimensional graded algebra generated in degree 1 over the finite-dimensional semisimple algebra $S = A_0$.

i. Fix a bimodule ${}_S V_S$ and an S-bimodule isomorphism $\varphi : V \xrightarrow{\sim} A_1$. Prove that

$$h_A(t) = h_{T_S(V)}(t) \quad \Longrightarrow \quad T_S(V) \xrightarrow{\sim} A,$$

where the isomorphism of graded algebras on the right-hand side is induced by id_S and φ.

ii. Consider $\Bbbk[x,z]/(x^2)$ with $\deg x = 1$ and $\deg z = 2$. Show that

$$h_B(t) = h_{T_{\Bbbk}(B_1)}(t) \qquad \text{but} \qquad B \ncong T_{\Bbbk}(B_1)$$

as graded algebras.

(3) For $q \in \Bbbk$ let

$$C_q = \Bbbk\langle x, y\rangle/(x^2,\ y^2,\ xy + qyx), \qquad \deg x = \deg y = 1.$$

i. Find a homogeneous basis of C_q and compute all products of two basis elements, treating $q = 0$ and $q \neq 0$. Justify linear independence of your basis.

ii. For $u = \alpha x + \beta y$ compute u^2, and determine

$$\{(\alpha, \beta) \in \Bbbk^2 \mid u^2 = 0\} \qquad \text{for each } q,$$

and the q for which every $u \in (C_q)_1$ satisfies $u^2 = 0$.

iii. Prove that

$$C_q \cong C_{q'} \ \text{ as graded algebras} \quad \Longleftrightarrow \quad q = q' \ \text{ or } \ qq' = 1.$$

Deduce that $C_1 \ncong C_{-1}$ if $\mathrm{char}\,\Bbbk \neq 2$ (even ungraded), and say what changes if $\mathrm{char}\,\Bbbk = 2$.
Hint. A graded isomorphism is determined by a linear bijection of the degree-1 parts, and preserves $\{\Bbbk u \mid u^2 = 0\}$.

iv. Compute a quadratic presentation of $C_q^!$ using θ_r (with $X = x^*$, $Y = y^*$), and $h_{C_q^!}(t)$. Verify

$$h_{C_q}(t)\, h_{C_q^!}(-t) = 1,$$

and compare $C_0^!$ with $C_1^!$.
Hint. Evaluate a general combination of X^2, XY, YX, Y^2 on x^2, y^2, and $xy + qyx$.